

Dynamic growth model for I-214 poplar plantations in the northern and central plateaux in Spain

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Received 14 December 2006; received in revised form 10 September 2007; accepted 13 October 2007

Abstract

Information required by forest managers about homogeneous, even-aged, single-species stands can be provided by whole-stand models, which are easily built with data often available in forest inventories and which represent a good compromise between generality and accuracy. The objective of the present study was to develop a dynamic growth model for I-214 clonal poplar plantations in the northern and central plateaux in Spain by use of data obtained from a network of 198 experimental plots measured between 2 and 12 times. The overall model is a hierarchical system that consists of three main components: a stand height projection function, a stand basal area projection function and a merchantable stand volume equation. In the model, the initial stand conditions at any point in time are defined by two state variables usually collected from common forest inventories: stand mean height and stand basal area. The model includes two transition functions, derived by the generalized algebraic difference approach, to enable projection of these variables at any particular time. The projected variables were incorporated into a merchantable stand volume equation to enable calculation of the commercial stand volume at a minimum diameter, which establish a merchantability threshold. Sensitive analysis of the accuracy of the predictions of the submodels according to the length of the time interval was achieved by use of the critical error statistic and the RMSE in relative proportion of the mean. Results indicated that the overall model provides satisfactory predictions for time intervals of between 1 and 4 years.

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Keywords: *Populus x euramericana*; I-214; Generalized algebraic difference approach; Dynamic growth model; Spain

1. Introduction

The wide range of forest growth models available differs in complexity and the detail in which they describe the systems under consideration. At one end of the range are the traditional empirical models based on periodic tree measurements, which make no attempt to measure all factors that may affect tree growth (Bruce and Wensel, 1988), and at the other end are the complex process models based on the mechanisms that underlie growth, which incorporate a large amount of information into the response functions (Johnsen et al., 2001). Korzukhin et al. (1996) reviewed the question of process-based or mechanistic,

as opposed to empirical, models. These authors concluded that process models offer significant advantages over empirical models for predicting and increasing our understanding of forest behaviour. These models are responsive to changes in environmental and site conditions and provide greater flexibility, generality and predictive power. On the other hand, empirical growth models do little to elucidate the mechanisms of tree or stand growth, and lack the ability to make predictions under an environmental gradient and changing environmental conditions. Nevertheless, they are currently widely used as practical tools in forest management, perhaps because of their simplicity.

Some authors (e.g., Mäkelä et al., 2000; Landsberg, 2003; Valentine and Mäkelä, 2005; Pinjuv et al., 2006) argue that the future lies in mixed or hybrid models that combine biophysical processes and relationships based on tree or stand-level measurements. They noted that if decision-making and analysis

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in forest management are to move towards a more general causal-oriented approach, mutual appreciation of methods and approaches by ecophysiological and empirical modellers is essential.

The choice of the most appropriate type of model depends on the purpose of its application, the stand characteristics, the resources available and the projection length (Vanclay, 1994; Burkhardt, 2003).

Of the two basic types of empirical models (whole-stand and individual-tree models), individual-tree models require a large number of state vectors, which will probably contain mostly redundant information, with a resulting loss of information (overparametrization); measuring the vectors may also be costly or the values will be impossible to obtain accurately from routine field sampling (García, 1994). Such models will probably be needed in more complex situations: uneven-aged and/or mixed species stands, or stands planted in rows or other patterns used in agroforestry (García, 1993). When dealing with homogeneous even-aged pure stands, whole-stand models are generally recommended (García, 1988, 1993; Vanclay, 1994), because they can be constructed on the basis of variables often available in forest inventory data, and they represent a good compromise between generality and accuracy of the estimates (Diéguez-Aranda et al., 2006b). Whole-stand models characterize the state of the stand by means of a small number of aggregate variables, such as basal area, mean diameter, volume per hectare, stems per hectare, average spacing, top height, etc. (García, 1993). When the aggregate variables are only available from a single initial inventory covering a wide range of ages, densities and sites, a static model may be developed. However, when more inventories are available, the corresponding data regarding the trajectories of the aggregate variables allow development of a dynamic growth model, which can be used to predict the rate of change of any stand variable. Integration of these equations (or summation when using discrete time) provides better description of the development of the stand over time than static models (García, 1988).

Intensively cultured poplar plantations are important in Spain and make significant contributions to industrial supplies of wood. At present there exist approximately 132,000 ha of poplar plantations in Spain (MAPA, 2005), which provide an average yield of $12 \text{ m}^3 \text{ ha}^{-1} \text{ year}^{-1}$ (Hernanz, 2001). The most widely used clone is 'I-214' *Populus x euramericana* (Dode) Guinier, which represents 70% of the poplar plantations in Castilla y León (Rueda, 2001), the most important region for poplar cultivation in Spain.

Traditionally, poplar plantations have been established along rivers valleys, and therefore compete with agriculture and other human activities for the soils; the methods of culture almost always involve ploughing the land several each year and the use of fertilization and irrigation when the trees are not deeply planted. The type of plantation has become attractive for farmers because of the high growth rates and short rotation periods.

Despite the high growth rate and the large surface area occupied by poplar plantations, to date very few studies have been carried out concerning growth and yield. The study of Bravo et al. (1996) is the only one studying which the growth

and yield of this species is investigated. Madrigal et al. (1999) presented yield tables for the Duero river basin based on the results of the former study. Other studies refer mainly to partial aspects of growth and yield or to specific locations (e.g., Padró, 1982; Padró and Hernández, 1983; Rodríguez and Blanco, 2001; Rodríguez, 2005).

The objective of the present study was to develop a dynamic whole-stand model for 'I-214' poplar clone plantations in the northern and central plateaux in Spain to help in decision-making regarding sustainable forest management.

2. Material and methods

2.1. Data

The data used to develop the whole-stand model were obtained from 198 experimental plots installed by the Spanish National Institute of Agricultural Research (INIA) in the regions of Castilla y León and Madrid at the beginning of the 1970s. The initial aim of establishing the network of experimental plots was to enable investigation of the effect of different cultural practices on growth and yield. However, since the plots were located throughout the area to cover a broad range of silvicultural states and levels of yield, they can be considered as adequate material for developing growth models. This database allows consideration of different treatment effects and it is therefore reasonable to assume that the effects will be embedded in the model.

The plot size varied from 400 to 640 m^2 , depending on the stand density, to include 16 trees per plot (plantation spacing ranged from 5 m $\times$ 5 m to 8 m $\times$ 5 m).

Trees (2-year-old plants with roots) were deep-planted (1.8–2 m). This planting procedure is widely used in Spain, and has been applied in more than 70% of the current plantations in Castilla y León (Rueda, 2001).

Two different sources of data were available in developing the whole-stand model: tree-by-tree plot data and average plot data. To obtain the former, the diameter and height of all the trees in each sample plot were measured. The diameter at breast height (1.3 m aboveground level) was measured twice (the second measurement at right angles to the first) with calipers, and total height was measured with a hypsometer. Total and merchantable volume of each tree in each inventory was calculated by the following volume ratio system developed by Barrio et al. (2006b) on the basis of data collected in some of the plots used in this study:

$$v_i = (-0.00115 + 0.000031d^2h) \exp\left(-0.33228 \frac{d_i^{4.963407}}{d^{4.492885}}\right) \quad (1)$$

where v_i (m^3) is the tree merchantable volume to an upper limit diameter d_i (cm), d (cm) is the diameter at breast height, and h (m) is the total tree height. Aggregation merchantable volume to upper limit diameters ranging between 0 and 30 cm of the plot trees allowed the stand merchantable volume to be obtained. In the second source of data only the following stand

variables were available: arithmetic mean diameter, stand mean height and number of trees per hectare.

Tree-by-tree plot data (108 plots) was used to develop all the submodels of the growth model, but average plot data (90 plots) could only be used to develop the stand height growth model because the arithmetic mean diameter was recorded rather than quadratic mean diameter. All the plots were measured between 2 and 12 times.

The following stand variables were computed for each plot and inventory: age (t , years), average height (H , m), stand basal area (G , $\text{m}^2 \text{ha}^{-1}$), quadratic mean diameter (dg , cm), number of stems per hectare (N), total (V , $\text{m}^3 \text{ha}^{-1}$) and commercial stand volume (V_c , $\text{m}^3 \text{ha}^{-1}$).

The mean, maximum, minimum values and standard deviations for each of the main stand variables calculated from the different inventories available are shown in Table 1.

2.2. Model structure

The proposed stand growth model is based on the state-space approach (Vancley, 1994), which makes use of state variables to characterize the system at an initial stage, and transition functions to project all or some of the state variables in the future.

In selecting the state variables, the principle of parsimony must be taken into account (Burkhart, 2003), i.e. the model should be the simplest one that describes the biological phenomena and that remains consistent with the structure and function of the actual biological system (Milsum, 1966). For unthinned stands, a two dimensional vector including dominant height and stand basal area as explanatory variables may be sufficient to describe the state of the stand at a given time (Pienaar and Turnbull, 1973). The inclusion of the number of trees (N) as an additional state variable was not considered since mortality is unimportant in intensively managed stands (García, 1993). Moreover, in clonal plantations of poplar in which the trees are widely spaced, the competition-related mortality can be considered negligible. Therefore, the status of the stands was characterized by two state variables: stand height (H) and stand basal area (G).

Two transition functions are used to project the two mentioned state variables at any point in time. These transition

functions must possess some important properties to provide consistent estimates: (i) consistency, i.e. no change for zero elapsed time; (ii) path-invariance, i.e. the result of projecting the state first from t_0 to t_1 , and then from t_1 to t_2 , must be the same as that of the one-step projection from t_0 to t_2 and (iii) causality, i.e. a change in status can only be influenced by inputs within the relevant time interval (García, 1994). Transition functions generated by integration of differential equations (or summation of difference equations when using discrete time) automatically satisfy these conditions.

The outputs of these two dynamic equations provide the inputs for the third submodel, which predicts the total or merchantable volume of the stand for a given age. Growth is subsequently calculated by subtraction.

2.2.1. Stand height and stand basal area transition functions

Stand height and stand basal area growth models are key components of whole-stand models, since they are directly related to economic variables such as stand volume.

Stand height of the largest trees is widely used to determine site quality, although it may be slightly affected by variations in stand density and intermediate cuttings (Monserud, 1988; Lloyd and Jones, 1982; Smith and Strub, 1991; Meridieu et al., 2003). Nevertheless, Usta (1983, cited in Birler, 1985) did not find any relationship between spacing and stand height growth for this clone when a space of more than 12 m^2 was available per tree. Moreover, in 'I-214' poplar plantations thinning is not regarded as economically viable and the plantations are widely spaced at establishment. Therefore, the mean height of the poplar I-214 plantations may be considered as a good indicator of site productivity and will be used in the present study.

Fulfillment of the above-mentioned properties for the transition functions depends on both the construction method and the mathematical function used to develop the model. Most of the properties can be achieved by use of techniques for dynamic equation derivation known in forestry as the algebraic difference approach (ADA) (Bailey and Clutter, 1974) or its generalization (GADA) (Cieszewski and Bailey, 2000).

Dynamic equations have the general form (omitting the vector of model parameters) of $Y=f(t, t_0, Y_0)$, where Y is the value of the function at age t , and Y_0 is the reference variable defined as the value of the function at age t_0 . The ADA essentially involves replacing a base-model site-specific parameter with its initial-condition solution. The GADA allows expansion of the base equations according to various theories about growth characteristics (e.g., asymptote, growth rate), thereby allowing more than one parameter to be site-specific and allowing the derivation of more flexible dynamic equations (see Cieszewski, 2001, 2002, 2003).

The first step in the ADA or GADA is to select a base equation and identify in it any desired number of site-specific parameters (only one parameter in ADA). An explicit definition of how the site-specific parameters change across different sites must be provided by replacing the parameters with explicit functions of X (one unobservable independent variable that describes site productivity as a summary of management

Table 1
Summary statistics of the data set used for model development

Stand variable	Mean	Minimum	Maximum	Standard deviation
t	6.40	2.00	13.00	2.87
H	16.38	2.42	28.20	5.53
G	13.88	2.12	26.95	6.80
N	264.97	250.00	505.05	21.10
dg	25.91	10.78	37.36	7.05
V	102.85	1.34	295.83	79.73

t , stand age (years); H , average stand height (m); G , stand basal area ($\text{m}^2 \text{ha}^{-1}$); N , number of stems per hectare; dg , quadratic mean diameter (cm); V , total stand volume ($\text{m}^3 \text{ha}^{-1}$).

regimes and ecological factors) and new parameters. In this way, the initially selected two-dimensional base equation ($Y=f(t)$) expands into an explicit three dimensional site equation ($Y=f(t,X)$) that describes both cross sectional and longitudinal changes with two independent variables t and X . Since X cannot be reliably measured or even functionally defined, the final step involves the substitution of X by equivalent initial conditions t_0 and Y_0 $Y=f(t, t_0, Y_0)$, so that the model can be implicitly defined and practically useful (Cieszewski and Bailey, 2000; Cieszewski, 2002).

The ADA or GADA can be applied in modelling the growth of any site dependent variable involving the use of unobservable variables substituted by the self-referencing concept (Northway, 1985) of model definition (Cieszewski, 2004), such as stand height, number of trees per unit area, stand basal area, stand volume, stand biomass or stand carbon sequestration.

In the present study, both recently developed dynamic equations with two site-specific parameters and frequently used dynamic equations with only one site-specific parameter were tested. We focused on the six dynamic equations used by Barrio et al. (2006a) for modelling the basal area growth of maritime pine in northwestern Spain. These equations are derived from three well-known growth functions (Korf—cited in Lundqvist, 1957, Hossfeld—Hossfeld 1822, and Bertalanffy-Richards—Bertalanffy, 1949, 1957; Richards, 1959) and have been widely used in modelling stand height and stand basal area (e.g., McDill and Amateis, 1992; Amaro et al., 1997; Falcão, 1997; Tomé et al., 2001; Cieszewski, 2002; Krumland and Eng, 2005; Diéguez-Aranda et al., 2006a,b; Barrio-Anta et al., 2007).

The formulations of the equations analyzed (T1–T6) are shown in Table 2. Following general notational convention, $a_1, a_2, \dots, a_n$ were used to denote parameters in base models, whereas $b_1, b_2, \dots, b_m$ were used for global parameters in subsequent GADA formulations.

2.2.2. Total and merchantable volume estimation

Estimation of the merchantable stand volume is required to achieve more efficient use of timber resources. Equations that allow determination of volume to a minimum diameter or height and thereby establish a merchantability threshold are known as merchantable volume equations. The volume ratio approach introduced by Burkhart (1977) has often been used in developing such equations (e.g., Cao et al., 1980; Clutter, 1980; Van Deusen et al., 1981; Reed and Green, 1984; Gregoire and Schabenberger, 1996; Valdez and Lynch, 2000; Tomé et al., 2001; Teshome, 2005; Barrio-Anta et al., 2007). Prediction of merchantable volume on the basis of this approach requires fitting a total stand volume equation (V) and a stand volume ratio equation (R_i). The customary approach in developing these models has been to fit both models separately (Avery and Burkhart, 1994). However, as Gregoire and Schabenberger (1996) have pointed out, it seems more reasonable to include both terms (R_i and V) in the same expression, in which case total volume becomes a special case of the volume ratio equation when the top diameter limit is equal to zero. Therefore,

Table 2
Base models and GADA formulations considered

Base equation	Parameter related to site	Solution for X with initial values (t_0, Y_0)	Dynamic equation	Model
Korf: $Y = a_1 \exp(-a_2 t^{-a_3})$	$a_2 = X$	$X_0 = -\ln\left(\frac{Y_0}{a_1}\right) t_0^{a_3}$	$Y = b_1 \left(\frac{Y_0}{b_1}\right)^{\frac{b_3}{b_1}} + e_{ij}$	(T1)
	$a_1 = \exp(X)$ $a_2 = b_1 + b_2/X$	$X_0 = \frac{1}{2} t_0^{-b_3} \left(b_1 + \frac{b_3}{t_0} \ln(Y_0) \pm \sqrt{4b_2 b_3} \pm \sqrt{4b_2 b_3 \ln(Y_0)} \right)^2$	$Y = \exp(X_0) \exp(-(b_1 + b_2/X_0) t^{b_3}) + e_{ij}$	(T2)
Hossfeld: $Y = \frac{a_1}{1+a_2 t^{-a_3}}$	$a_2 = X$	$X_0 = t_0^{-a_3} \left(\frac{a_1}{Y_0} - 1 \right)$	$Y = b_1 / (1 - (1 - b_1/Y_0)(t_0/t)^{b_3}) + e_{ij}$	(T3)
	$a_1 = b_1 + X$ $a_2 = b_2/X$	$X_0 = \frac{1}{2} \left(Y_0 - b_1 \pm \sqrt{(Y_0 - b_1)^2 + 4b_2 Y_0 t_0^{-b_3}} \right)$	$Y = \frac{b_1 + X_0}{1 + b_2/X_0 t^{b_3}} + e_{ij}$	(T4)
Bertalanffy-Richards: $Y = a_1 (1 - \exp(-a_2 t))^{a_3}$	$a_2 = X$	$X_0 = -\ln(1 - (Y_0/b_1)^{1/b_3}) / t_0$	$Y = b_1 \left(1 - \left(1 - \left(\frac{Y_0}{b_1} \right)^{1/b_3} \right)^{t/t_0} \right)^{b_3} + e_{ij}$	(T5)
	$a_1 = \exp(X)$ $a_3 = b_2 + b_3/X$	$X_0 = \frac{1}{2} \left(\ln Y_0 - b_2 L_0 + \sqrt{(\ln Y_0 - b_2 L_0)^2 - 4b_3 L_0} \right)$ with $L_0 = \ln(1 - \exp(-b_1 t_0))$	$Y = Y_0 \left(\frac{1 - \exp(-b_1 t)}{1 - \exp(-b_1 t_0)} \right)^{(b_2 + b_3/X_0)}$	(T6)

Table 3
Merchantable stand volume equations considered

Total volume equation (V)	Ratio equation (R_i)	Composite model ($V_i = V \cdot R_i$)	Model
$V = b_1 G^{b_2} H^{b_3}$	$R_i = 1 + b_4 \left(\frac{d_i^{b_5}}{dg^{b_6}} \right)$	$V_i = b_1 G^{b_2} H^{b_3} \left(1 + b_4 \left(\frac{d_i^{b_5}}{dg^{b_6}} \right) \right) + e_{ij}$	(V1)
	$R_i = \exp(b_4 d_i^{b_5} dg^{b_6})$	$V_i = b_1 G^{b_2} H^{b_3} \exp(b_4 d_i^{b_5} dg^{b_6}) + e_{ij}$	(V2)
	$R_i = 1 + \left(\frac{b_4 d_i^{b_5}}{dg^{b_6} H^{b_7}} \right)$	$V_i = b_1 G^{b_2} H^{b_3} \left(1 + \left(\frac{b_4 d_i^{b_5}}{dg^{b_6} H^{b_7}} \right) \right) + e_{ij}$	(V3)

V , total stand volume ($\text{m}^3 \text{ha}^{-1}$); V_i , merchantable stand volume ($\text{m}^3 \text{ha}^{-1}$); G , stand basal area ($\text{m}^2 \text{ha}^{-1}$); H , average stand height (m); d_i , top diameter limit (cm); dg , quadratic mean diameter (cm); b_i ($i = 1-7$), parameters to be estimated.

composite models including both total volume and volume ratio equations were selected for analysis.

The commonly used allometric model, which includes stand height and stand basal area as independent variables, was selected to determine the total stand volume. Three different models, which depend on the stand quadratic mean diameter and on the top diameter limit, were initially considered as volume ratio equations. The first equation (V1) was proposed by [Burkhart \(1977\)](#). The second equation (V2) is a variation of the exponential ratio model originally developed by [Van Deusen et al. \(1981\)](#), modified by adding a third parameter ([Clark and Thomas, 1984](#); see also [Tasissa et al., 1997](#)). The third equation (V3) is a modified version of the model proposed by [Reed and Green \(1984\)](#), which includes the stand height as an additional variable. In all of the original models the tree variables were replaced by the equivalent stand-related variables (e.g., tree height was replaced by stand height and diameter at breast height by quadratic mean diameter). The formulations of each ratio equation and the composite models are shown in [Table 3](#).

2.3. Model fitting

The individual trends represented in stand height and stand basal area data of the plots can be modelled by considering that individual responses all follow a similar functional form with parameters that vary among individuals (local parameters) and parameters that are common for all individuals (global parameters). In practice both base-age specific (BAS) and base-age invariant (BAI) methods can be used. The assumption behind the BAS methods, which use selected data (e.g., heights at the given base-age) as site-specific constants, is that the data measurements simultaneously do and do not contain measurement and environmental errors (on the left- and right-hand sides of the model, respectively) (e.g., [Curtis et al., 1974](#); [Monserud, 1988](#)) which is clearly untenable ([Bailey and Clutter, 1974](#)). The assumption underlying the BAI methods, which estimate site-specific effects, is that the data measurements always contain measurement and environmental errors (on both the left- and right-hand sides of the model), which must be modelled ([Cieszewski, 2003](#)). From among the different BAI parameter estimation techniques, we estimated the random site-specific effects simultaneously with the fixed effects using the dummy variables method described by [Cieszewski and Bailey \(2000\)](#). In this method the initial conditions are specified as

identical for all the measurements belonging to the same tree or plot. The initial age can be selected arbitrarily, within limits, although age zero is not allowed. The stand variables corresponding to the initial age are then simultaneously estimated for each tree or plot along with all of the global model parameters during the fitting process. The dummy variables method recognizes that each measurement is made with error and therefore it does not force the model through any given measurement. Rather, the curve is fitted to the observed individual trends in the data. With this method all of the data can be used, and there is no need to make any arbitrary choice regarding measurement intervals.

In the general formulation of the dynamic equations, the error terms e_{ij} are assumed to be independent and identically distributed with zero mean. Nevertheless, because of the longitudinal nature of the data sets used for model fitting, correlation between the residuals within the same individual (plot or tree) may be expected. To overcome any possible autocorrelation, we modelled the error terms using a continuous-autoregressive error structure (CAR(x)), which allows the model to be applied to irregularly spaced, unbalanced data ([Gregoire et al., 1995](#); [Zimmerman and Núñez-Antón, 2001](#)). To account for k -order autocorrelation, a CAR(x) model form that expands the error terms in the following way can be used ([Zimmerman and Núñez-Antón, 2001](#)):

$$e_{ij} = \sum_{k=1}^{k=x} I_k \rho_k^{t_{ij}-t_{ij-k}} e_{ij-k} + \varepsilon_{ij} \quad (2)$$

where e_{ij} is the j th ordinary residual on the i th individual, e_{ij-k} is the $j-k$ th ordinary residual on the i th individual, $I_k = 1$ for $j > k$ and it is zero for $j \leq k$, ρ_k is the k -order autoregressive parameter to be estimated, and $t_{ij} - t_{ij-k}$ is the time (or the distance— $h_{ij} - h_{ij-k}$ —in merchantable volume equation) that separates the j th from the $j-k$ th observations within each individual. In such cases ε_{ij} now includes the error terms under conditions of independence.

To evaluate the presence of autocorrelation and the order of the CAR(x) to be used, graphs representing residuals plotted against lag-residuals from previous observations within each tree or plot were examined visually. The dummy variables method and the CAR(x) error structure was programmed using the SAS/ETS[®] MODEL procedure ([SAS Institute Inc., 2004](#)), which allows for dynamic updating of the residuals.

The best model for estimating stand height and stand basal area growth was selected by application of the criteria described in the following section. Both equations were then fitted simultaneously to minimize the total sum of squared errors of the system (both stand height and basal area prediction errors) and to take into account the correlation between residuals of both variables. Since different amounts of data for height and basal area were available, an associated dummy variable was included for each dependent variable to specify the number of data and to allow use of all the data when fitting the equations simultaneously.

The simultaneous fitting was accomplished by use of the *seemingly unrelated regression* (SUR) technique (Zellner, 1962) implemented in the SAS/ETS[®] MODEL procedure (SAS Institute Inc., 2004), and involved the use of the dummy variables method and the CAR(x) error structure.

2.4. Model comparison and evaluation

Comparison of the estimates for the different models fitted was based on numerical and graphical analyses of the residuals. Two statistics were examined: the root mean square error (RMSE) and the coefficient of determination for nonlinear regression (R^2). The expressions of these statistics are as follows:

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n - p}} \quad (3)$$

$$R^2 = r_{y_i \hat{y}_i}^2 \quad (4)$$

where y_i and $\hat{y}_i$ are the observed and predicted values of the dependent variable, respectively, n the total number of observations and p the number of model parameters; $r_{y_i \hat{y}_i}$ is the correlation coefficient for a linear regression between the observed and predicted values of the dependent variable (see Ryan, 1997, pp. 419 and 424).

Other important steps in evaluating the models were graphical analysis of the residuals and examination of the appearance of the fitted curves overlaid on the trajectories of the dependent variables for each plot. Visual or graphical inspection is an essential point in selecting the most appropriate model because curve profiles may differ drastically, even though fit statistics and residuals are similar (Huang et al., 2003; Diéguez-Aranda et al., 2006a; Barrio et al., 2006a).

Once the submodels have been constructed, assessment of their validity is often needed to ensure that the predictions represent the most likely outcome in the real world (Yang et al., 2004). Moreover, although the behaviour of individual submodels within the model plays an important role in determining the overall outcome, the validity of each individual component does not guarantee the validity of the overall outcome, which is usually considered more important in practice. The overall model outcome must therefore also be evaluated. The only method that can be regarded as “true” validation involves the use of a new independent data set (Bruce and Wensel, 1988; Vanclay and Skovsgaard, 1997; Pretzsch et al., 2002; Yang et al., 2004; Jordan et al., 2006) but the scarcity of such data forces the use of alternative approaches. The common method of splitting the data set in two portions does not provide additional information (Huang et al., 2003) and neither is recommended from the point of view of parameter estimation (Myers, 1990; Hirsch, 1991). Moreover, other techniques such as double cross-validation or statistical testing provide very limited information about the predictive ability of the models (Kozak and Kozak, 2003; Yang et al., 2004). Therefore, we decided not to attempt a model validation in this study, and relied on the fact that a well-developed model will behave well. The submodels were evaluated to confirm their usefulness and reliability by the goodness-of-fit statistics and the graphical analysis described above. A sensitive analysis was also carried out to obtain some

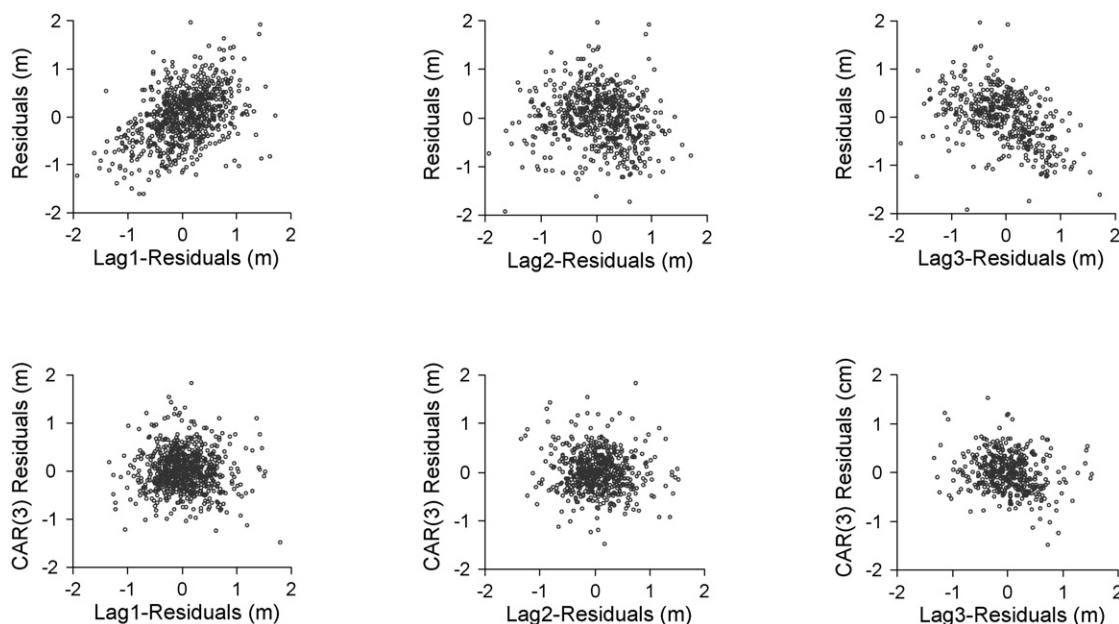


Fig. 1. Residuals versus age-lag1-residuals (left column), age-lag2-residuals (middle column) and age-lag3-residuals (right column) for stand height projection function without considering autocorrelation parameters (first row), and using a third-order autoregressive error structure (second row).

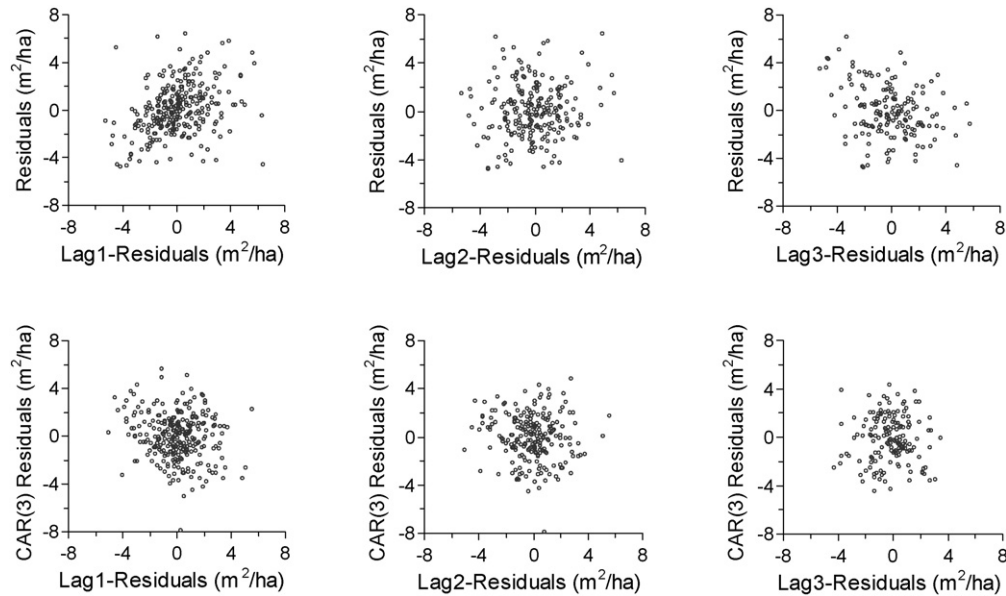


Fig. 2. Residuals versus age-lag1-residuals (left column), age-lag2-residuals (middle column) and age-lag3-residuals (right column) for basal area projection function without considering autocorrelation parameters (first row), and using a third-order autoregressive error structure (second row).

information about the predictive ability of the submodels and the overall whole-stand model. For this, observed state variables from any one inventory of the plots were projected forward for different time intervals using the stand height and stand basal area dynamic equations. All possible non-descending growth intervals were therefore considered in this analysis. In addition, the values obtained were used to estimate total stand volume at different projection intervals including the three components of the whole-stand model. It must be taken into account that total stand volume is the critical output variable of the whole model, since its estimation involves all the functions included and is important in economic assessments.

To evaluate the most appropriate time interval for projections, we used the critical error ($E_{crit.}$). The critical error is expressed as a percentage of the observed mean and is computed by re-arranging Freese's χ_n^2 statistic (Reynolds, 1984; Robinson and Froese, 2004):

$$E_{crit.} = \frac{\sqrt{\tau^2 \sum_{i=1}^n (y_i - \hat{y}_i)^2 / \chi_{crit.}^2}}{\bar{y}} \quad (5)$$

where n is the total number of observations in the data set, y_i is the observed value, $\hat{y}_i$ the value predicted from the fitted model,

$\bar{y}$ is the average of the observed values, τ is a standard normal deviate at the specified probability level ($\tau = 1.96$ for $\alpha = 0.05$), and $\chi_{crit.}^2$ is obtained for $\alpha = 0.05$ and n degrees of freedom. If the specified allowable error expressed as a percentage of the observed mean is within the limit of the critical error, the χ_n^2 test indicates that the model does not give satisfactory predictions; the contrary result indicates that the predictions are acceptable.

To determine the time interval (lag) that provides acceptable errors, the root mean squared error and critical error ($E_{crit.}$) for stand height, stand basal area and total stand volume predictions were plotted against different time intervals.

3. Results and discussion

3.1. Transition functions for stand height and stand basal area

Initially only the mean structure of the data was fitted and a trend in residuals as a function of age-lag-residuals within the same plot was observed in all the models considered. This result was expected because of the longitudinal nature of the data used for both stand height and stand basal area growth modelling.

Table 4

Goodness-of-fit statistics for the stand height and stand basal area dynamic equations and for the merchantable stand volume equation

Model	Stand height projection function ($n = 1678$)		Basal area projection function ($n = 804$)		Merchantable stand volume equation ($n = 14895$)		
	R^2	RMSE	R^2	RMSE	Model	R^2	RMSE
T1	0.9925	0.5150	0.9951	0.6384	V1	0.9998	0.6515
T2	0.9925	0.5147	0.9956	0.6030	V2	0.9999	0.3342
T3	0.9929	0.5023	0.9952	0.6345	V3	0.9998	0.6493
T4	0.9929	0.5022	0.9960	0.5739			
T5	0.9929	0.5004	0.9953	0.6302			
T6	0.9927	0.5071	0.9960	0.5764			

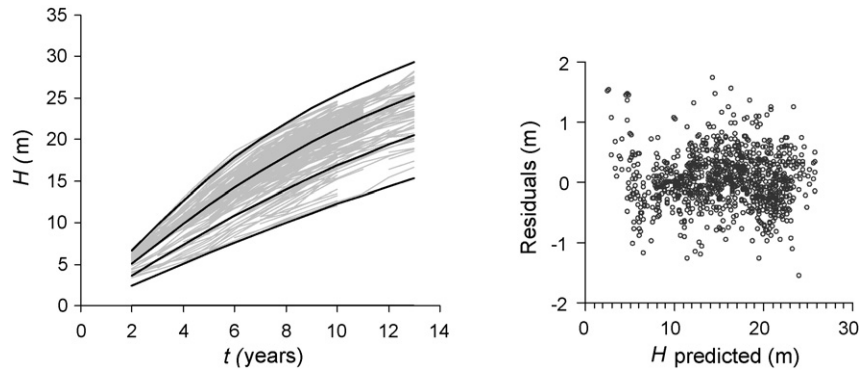


Fig. 3. Average stand height prediction curves for initial values of 10, 14, 18 and 22 m at 8 years overlaid on the trajectories of the observed values over time (left-hand side) and pattern of residuals versus predicted values (right-hand side).

A third-order continuous-time autoregressive error structure was required for modelling the inherent autocorrelation of the hierarchical data. The results showed that all of the parameters associated with the error term were significant at a probability level of 0.05. After correction for autocorrelation, the trends in the graphs of residuals as a function of age-lag-residuals disappeared, as inferred from Figs. 1 and 2 and from the Durbin-Watson statistic (value of about 2 for all models).

As regards the stand height growth model, all the equations considered achieved convergence and were found to be significant at $P < 0.05$; however, three models in which two parameters were considered to be site-specific showed some non significant parameters. Among all the equations tested, model T5, derived from the Bertalanfy-Richards function solved by parameter a_2 provided the best results for the statistics used for comparison (Table 4). This model also showed adequate graphical behaviour, as it produced the most adequate height growth curves (Fig. 3, left-hand side) with a random pattern of residuals around zero with homogeneous variance and no discernable trend (Fig. 3, right-hand side). Model T5 was therefore selected for developing the productivity curves for 'I-214' poplar stands.

Although site quality and site productivity refer to same thing, conceptually they are different. Thus, site quality refers to the inherent capacity of forest land to sustain vegetative growth (Clutter et al., 1983). However, this inherent capacity can be modified by altering the system as

occurs in the poplar plantations in which irrigation, fertilization and tilling are common carried out. When such modifications are considered in assessing the potential productivity of the site, the corresponding term is site productivity. The stand height growth curves developed here must therefore be considered as productivity curves rather than site quality curves.

Comparison among stands requires a productivity index equivalent to the common site index, which requires the selection of a reference age. For this, different base ages and their corresponding observed heights were used to estimate heights at other ages for each plot. The results were compared with the values obtained by remeasuring plots, and the relative

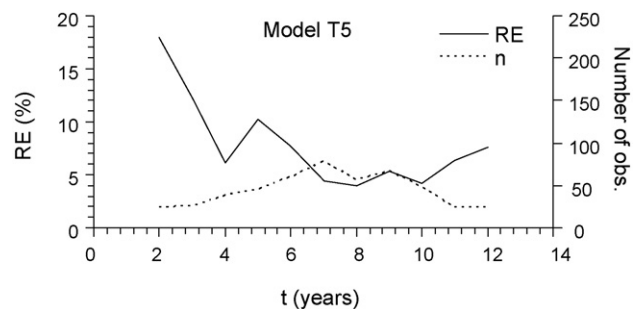


Fig. 4. Relative error in stand height prediction related to the choice of a reference age for stand height model T5.

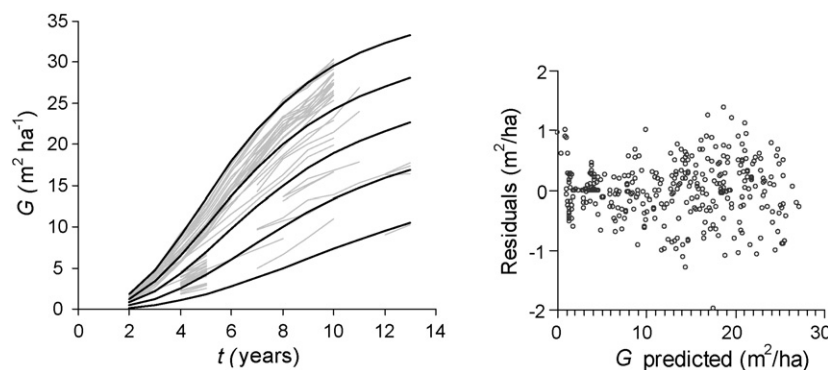


Fig. 5. Basal area prediction curves for initial values of 20, 40, 60, 80 and 100 $\text{m}^2 \text{ha}^{-1}$ at 8 years overlaid on the trajectories of the observed values over time (left-hand side) and pattern of residuals versus predicted values (right-hand side).

error in predictions (RE%) was then calculated as follows:

$$\text{RE}\% = 100 \frac{\sqrt{\sum_{i=1}^n (H_i - \hat{H}_i)^2 / (n - p)}}{\bar{H}} \quad (6)$$

where H_i , $\hat{H}_i$ and $\bar{H}$ are the observed, predicted and average values of the stand height, respectively, n is the total number of observations used to fit the model and p is the number of parameters in the model.

Finally, a reference age of 8 years was found to be best for predicting height at other ages with a minimum degree of reliability (Fig. 4).

As regards the stand basal area growth model, dynamic equations with two site-specific parameters performed better than models with only one site-specific parameter. This may be because former models are more flexible than the latter and perform better when data are not well represented by qualities and ages as occur with the basal area database. Similar results were obtained by Barrio et al. (2006a) and Castedo-Dorado et al. (2007) in developing a basal area projection function for *Pinus pinaster* and *Pinus radiata* stands, respectively, in northwest Spain.

After considering statistical criteria and graphical analysis of the stand basal area growth curves overlaid on the trajectories of the observed values over time, model T4 was selected as the best compromise between biological behaviour and goodness-of-fit (Fig. 5, left-hand side). This model explained 99.6% of the total variation in data (Table 4) and provided a random pattern of residuals around zero with homogeneous variance (Fig. 5, right-hand side). As previously stated, simultaneous fitting of both stand height and basal area models was accomplished to minimize the total sum of squared errors in the system and to take into account the correlation between residuals of the two variables ($\text{Corr}(H, G) = 0.36$). The asymptotic 95% confidence intervals obtained for the estimated parameters from equation T4 and T5 showed that all were significant (Table 5). The accuracy of the stand height and basal area estimations, given by the root mean squared error, were 0.50 m and 0.57 m² ha⁻¹, which represent 3.0 and 4.7% of the observed means value, respectively.

3.2. Merchantable volume equation

Because of the hierarchical nature of the data there was strong autocorrelation among observations associated with the merchantable volume database. Model fitting in which third-order continuous-autoregressive error structure was considered only partially removed the problem. However, it must be taken into account that the sole purpose of using the autocorrelation structure was to obtain more efficient parameter estimates. Considering that all the parameters were highly significant (including the autocorrelation parameters) (Table 5), no changes in parameter significance would be achieved by more efficient incorporation of the residual trends into the model.

Among the three models analyzed, merchantable stand volume equation V2 performed best and was more accurate than the other two models (Table 4). This model can provide the

Table 5
Parameters estimates for the stand height and stand basal area dynamic equations and for the merchantable stand volume equation

Equation	Variable	Estimated coefficients	b_1	b_2	b_3	b_4	b_5	b_6	ρ_1	ρ_2	ρ_3
T5	H		39.275 (1.4116)	1.1121 (0.025)	—	—	—	—	0.61238 (0.0400)	0.55056 (0.0479)	-0.33284 (0.1434)
T4	G		12.944 (5.6461)	3242.1 (1260.7)	2.6389 (0.1335)	—	—	—	1.2106 (0.0772)	-1.0703 (0.0506)	0.69347 (0.1067)
V2	V_i		0.36281 (0.00076)	0.97834 (0.00049)	1.0506 (0.00098)	-0.32504 (0.00493)	4.7936 (0.00383)	-4.3337 (0.00507)	1.5665 (0.00594)	1.3827 (0.00823)	0.63488 (0.00597)

Values in parantheses are the standard errors of the estimated coefficients.

merchantable volume of a stand at a given time once the stand height and the stand basal area are known for that time. These state variables can be obtained from an ordinary forest inventory in which diameter at breast height and height are measured or can be easily projected over time by their corresponding transition functions. Quadratic mean diameter can be directly obtained by taking into account that the number of trees per hectare is considered to be constant over time.

3.3. Overall evaluation of the model

Considering the required accuracy in forestry growth modelling, in which a mean prediction error of the observed mean at 95% confidence intervals within ± 10 –20% is generally realistic and reasonable as a limit for the actual choice of the acceptance and rejection levels (Huang et al., 2003), we can state that the stand height model provides satisfactory predictions for time intervals of 8 years. The accuracy of the

predictions for this age interval, given by the RMSE, was 3.02 m, which represents 18.5% of the observed mean value (Fig. 6).

Following the same criteria, basal area predictions for a time interval of 4 years showed an $E_{\text{crit.}}$ of 20.9%, and a RMSE of $2.5 \text{ m}^2 \text{ ha}^{-1}$, which represents 18.2% of the observed mean value (Fig. 6).

To evaluate the critical error in stand volume estimations, the projected values of stand height and stand basal area were estimated using equations P5 and P4, respectively. These values were used as independent variables to compute the total stand volume for different time intervals. As can be observed in Fig. 6, volume estimates should not be made directly for differences in ages of more than 4 years for not exceed an $E_{\text{crit.}}$ value of 20%. For this time interval, the RMSE of volume predictions was $15.9 \text{ m}^3 \text{ ha}^{-1}$, which represents 15.4% of the observed mean value (i.e if the predictions are made for a time interval of 8 years, the RMSE would represent 27.3% of the observed mean).

In summary, for the available dataset and considering a specified allowable error expressed as a percentage of the observed mean of 20%, the model was found to provide satisfactory global predictions for time intervals of between 1 and 4 years. For longer projection intervals, new inventory data should be used to establish a new starting point for estimations.

Acknowledgements

The authors gratefully acknowledge the dedication shown by Federico González Antoñanzas in the study of poplar plantations in Spain and for his efforts in collecting data such as that used in the present study. We thank Dr. Christine Francis for correcting the English grammar of the text.

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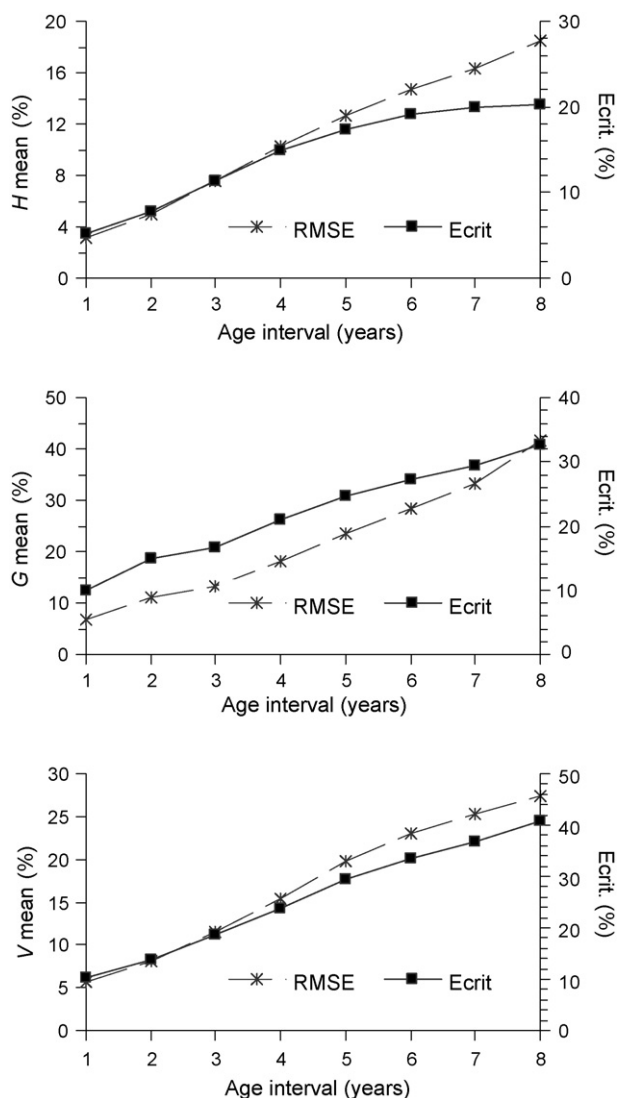


Fig. 6. Plots of critical error for stand height (H), stand basal area (G) and total stand volume (V) predictions against different year lags. Stand volume was calculated using the predicted values of stand height and stand basal area.

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